

type	L=1	L=3	L=5
$S_f^{\Sigma''}$	Overlap integrals (odd) - q_cutoff=1050.079	$6.7(26)_m \times 10^{-5}$	$1.707(24)_m \times 10^{-5}$
$S_f^{\Sigma''}$	$-1.6(25)_m \times 10^{-5}$	$3.74(36)_m \times 10^{-6}$	$-2.9(63)_m \times 10^{-7}$
$S_f^{\prime\prime, \Sigma''}$	$6.18(20)_m(6)_{ex} \times 10^{-4}$	$-8.2(32)_m \times 10^{-5}$	$2.296(29)_m(20)_{ex} \times 10^{-5}$
$S_f^{\prime\prime, \Sigma''}$	$1.8(28)_m(12)_{ex} \times 10^{-5}$	$-3.59(77)_m \times 10^{-6}$	$4(11)_m \times 10^{-7}$
$S_f^{(4), \Sigma''}$	$-8.9(4)_m(28)_{ex} \times 10^{-4}$	$1.12(64)_m(10)_{ex} \times 10^{-4}$	$-4.15(1)_m(12)_{ex} \times 10^{-5}$
$S_f^{(4), \Sigma''}$	$1.7(1)_m(15)_{ex} \times 10^{-4}$	$-3.2(44)_m(26)_{ex} \times 10^{-5}$	$-7(14)_m \times 10^{-7}$
$S_g^{\Sigma''}$	$5.11(13)_m \times 10^{-4}$	$-5.6(23)_m \times 10^{-5}$	$1.492(22)_m \times 10^{-5}$
$S_g^{\Sigma''}$	$1.7(24)_m \times 10^{-5}$	$-3.54(29)_m \times 10^{-6}$	$2.7(54)_m \times 10^{-7}$
$S_g^{\prime\prime, \Sigma''}$	$-5.74(20)_m(5)_{ex} \times 10^{-4}$	$6.9(27)_m \times 10^{-5}$	$-1.875(27)_m(17)_{ex} \times 10^{-5}$
$S_g^{\prime\prime, \Sigma''}$	$-2.0(25)_m(13)_{ex} \times 10^{-5}$	$3.60(56)_m \times 10^{-6}$	$-4.1(88)_m \times 10^{-7}$
$S_g^{(4), \Sigma''}$	$8.0(4)_m(29)_{ex} \times 10^{-4}$	$-8.0(53)_m(9)_{ex} \times 10^{-5}$	$2.94(2)_m(11)_{ex} \times 10^{-5}$
$S_g^{(4), \Sigma''}$	$-1.6(2)_m(14)_{ex} \times 10^{-4}$	$2.7(39)_m(33)_{ex} \times 10^{-5}$	$7.7(80)_m(3)_{ex} \times 10^{-7}$
$V_{f+}^{\Sigma''}$		$4.3(17)_m \times 10^{-5}$	$-1.152(17)_m \times 10^{-5}$
$V_{f+}^{\Sigma''}$		$2.51(26)_m \times 10^{-6}$	$-2.0(42)_m \times 10^{-7}$
$V_{f+}^{\prime\prime, \Sigma''}$	$-2.08(25)_m(5)_{ex} \times 10^{-4}$	$2.91(43)_m \times 10^{-5}$	$-3.82(13)_m(2)_{ex} \times 10^{-6}$
$V_{f+}^{\prime\prime, \Sigma''}$	$1.09(33)_m(67)_{ex} \times 10^{-4}$	$-5.27(87)_m \times 10^{-6}$	$1.9(25)_m \times 10^{-7}$
$V_{g+}^{\Sigma''}$	$3.329(81)_m \times 10^{-4}$	$-4.6(18)_m \times 10^{-5}$	$1.295(18)_m \times 10^{-5}$
$V_{g+}^{\Sigma''}$	$10(16)_m \times 10^{-6}$	$-2.63(28)_m \times 10^{-6}$	$2.2(48)_m \times 10^{-7}$
$V_{g+}^{\prime\prime, \Sigma''}$	$2.17(26)_m(5)_{ex} \times 10^{-4}$	$-3.10(44)_m \times 10^{-5}$	$3.87(16)_m(3)_{ex} \times 10^{-6}$
$V_{g+}^{\prime\prime, \Sigma''}$	$-1.12(34)_m(69)_{ex} \times 10^{-4}$	$5.84(92)_m \times 10^{-6}$	$-2.3(27)_m \times 10^{-7}$
$V_{f-}^{\Sigma''}$	$4.04(10)_m \times 10^{-4}$	$-3.9(16)_m \times 10^{-5}$	$9.82(15)_m \times 10^{-6}$
$V_{f-}^{\Sigma''}$	$1.4(19)_m \times 10^{-5}$	$-2.54(18)_m \times 10^{-6}$	$1.9(35)_m \times 10^{-7}$
$V_{f-}^{\prime\prime, \Sigma''}$	$-1.518(19)_m(3)_{ex} \times 10^{-3}$	$3.34(94)_m \times 10^{-4}$	$-8.773(14)_m(52)_{ex} \times 10^{-5}$
$V_{f-}^{\prime\prime, \Sigma''}$	$2.7(12)_m(5)_{ex} \times 10^{-4}$	$-2.08(53)_m \times 10^{-5}$	$1(48)_m \times 10^{-7}$
$V_{g-}^{\Sigma''}$	$-4.29(11)_m \times 10^{-4}$	$4.3(18)_m \times 10^{-5}$	$-1.106(16)_m \times 10^{-5}$
$V_{g-}^{\Sigma''}$	$-1.5(20)_m \times 10^{-5}$	$2.70(22)_m \times 10^{-6}$	$-2.0(40)_m \times 10^{-7}$
$V_{g-}^{\prime\prime, \Sigma''}$	$1.576(19)_m(3)_{ex} \times 10^{-3}$	$-3.6(10)_m \times 10^{-4}$	$1.0151(28)_m(58)_{ex} \times 10^{-4}$
$V_{g-}^{\prime\prime, \Sigma''}$	$-2.8(12)_m(5)_{ex} \times 10^{-4}$	$2.35(58)_m \times 10^{-5}$	$-3(56)_m \times 10^{-7}$
$V_{ffbar}^{\Sigma''}$	$1.350(31)_m \times 10^{-5}$	$-2.8(11)_m \times 10^{-6}$	$9.463(96)_m \times 10^{-7}$
$V_{ffbar}^{\Sigma''}$	$3.1(64)_m \times 10^{-7}$	$-1.09(27)_m \times 10^{-7}$	$1.2(38)_m \times 10^{-8}$
$V_{ffbar}^{\prime\prime, \Sigma''}$	$-1.160(19)_m(8)_{ex} \times 10^{-5}$	$1.43(49)_m \times 10^{-6}$	$-3.253(14)_m(22)_{ex} \times 10^{-7}$
$V_{ffbar}^{\prime\prime, \Sigma''}$	$1.8(64)_m(14)_{ex} \times 10^{-7}$	$7(22)_m \times 10^{-9}$	$-3(16)_m \times 10^{-9}$
$V_{gfbar}^{\Sigma''}$	$-1.348(32)_m \times 10^{-5}$	$2.8(11)_m \times 10^{-6}$	$-9.240(96)_m \times 10^{-7}$
$V_{gfbar}^{\Sigma''}$	$-3.4(63)_m \times 10^{-7}$	$1.11(25)_m \times 10^{-7}$	$-1.2(37)_m \times 10^{-8}$
$V_{gfbar}^{\prime\prime, \Sigma''}$	$1.137(20)_m(8)_{ex} \times 10^{-5}$	$-1.36(47)_m \times 10^{-6}$	$3.119(15)_m(22)_{ex} \times 10^{-7}$
$V_{gfbar}^{\prime\prime, \Sigma''}$	$-1.4(62)_m(11)_{ex} \times 10^{-7}$	$-1.1(20)_m \times 10^{-8}$	$3(16)_m \times 10^{-9}$
$T_{f+}^{\Sigma''}$		$5.2(20)_m \times 10^{-5}$	$-1.531(18)_m \times 10^{-5}$
$T_{f+}^{\Sigma''}$		$2.51(40)_m \times 10^{-6}$	$-2.2(59)_m \times 10^{-7}$
$T_{g+}^{\Sigma''}$	$3.257(71)_m \times 10^{-4}$	$-4.5(18)_m \times 10^{-5}$	$1.276(17)_m \times 10^{-5}$
$T_{g+}^{\Sigma''}$	$8(16)_m \times 10^{-6}$	$-2.45(30)_m \times 10^{-6}$	$2.1(48)_m \times 10^{-7}$
$T_{f-}^{\Sigma''}$	$4.40(10)_m \times 10^{-4}$	$-4.7(19)_m \times 10^{-5}$	$1.237(17)_m \times 10^{-5}$
$T_{f-}^{\Sigma''}$	$1.1(21)_m \times 10^{-5}$	$-2.69(29)_m \times 10^{-6}$	$2.1(46)_m \times 10^{-7}$
$T_{g-}^{\Sigma''}$	$-4.053(98)_m \times 10^{-4}$	$4.1(17)_m \times 10^{-5}$	$-1.070(16)_m \times 10^{-5}$
$T_{g-}^{\Sigma''}$	$-1.2(19)_m \times 10^{-5}$	$2.53(22)_m \times 10^{-6}$	$-1.9(39)_m \times 10^{-7}$
$T_{ffbar}^{\Sigma''}$	$1.350(31)_m \times 10^{-5}$	$-2.8(11)_m \times 10^{-6}$	$9.463(96)_m \times 10^{-7}$
$T_{ffbar}^{\Sigma''}$	$3.1(64)_m \times 10^{-7}$	$-1.09(27)_m \times 10^{-7}$	$1.2(38)_m \times 10^{-8}$
$T_{gfbar}^{\Sigma''}$	$-1.348(32)_m \times 10^{-5}$	$2.8(11)_m \times 10^{-6}$	$-9.240(96)_m \times 10^{-7}$
$T_{gfbar}^{\Sigma''}$	$-3.4(63)_m \times 10^{-7}$	$1.11(25)_m \times 10^{-7}$	$-1.2(37)_m \times 10^{-8}$
$V_f^{\prime\prime, \Sigma'}$	$5.40(32)_m \times 10^{-4}$	$-6.7(30)_m \times 10^{-5}$	$1.84(41)_{ex} \times 10^{-5}$
$V_f^{\prime\prime, \Sigma'}$	$1.55(41)_m \times 10^{-4}$	$-1.1(1)_m(26)_{ex} \times 10^{-5}$	$8.3(81)_m(19)_{ex} \times 10^{-7}$
$V_g^{\prime\prime, \Sigma'}$	$-5.62(33)_m \times 10^{-4}$	$7.3(33)_m \times 10^{-5}$	$-2.13(47)_{ex} \times 10^{-5}$
$V_g^{\prime\prime, \Sigma'}$	$-1.61(43)_m \times 10^{-4}$	$1.2(1)_m(12)_{ex} \times 10^{-5}$	$-9.7(94)_m(21)_{ex} \times 10^{-7}$
$V_{f-f}^{\prime\prime, \Sigma'}$		$6.9(25)_m \times 10^{-6}$	$-2.82(3)_m(50)_{ex} \times 10^{-6}$
$V_{f-f}^{\prime\prime, \Sigma'}$		$1.384(27)_m(40)_{ex} \times 10^{-6}$	$-1.4(13)_m(2)_{ex} \times 10^{-7}$
$V_{g-f}^{\prime\prime, \Sigma'}$	$1.140(61)_m(3)_{ex} \times 10^{-5}$	$-6.7(25)_m \times 10^{-6}$	$2.71(3)_m(48)_{ex} \times 10^{-6}$
$V_{g-f}^{\prime\prime, \Sigma'}$	$6.7(23)_m \times 10^{-6}$	$-1.324(29)_m(41)_{ex} \times 10^{-6}$	$1.3(12)_m(2)_{ex} \times 10^{-7}$
$V_{f+f}^{\prime\prime, \Sigma'}$	$-6.85(45)_m(1)_{ex} \times 10^{-6}$	$3.3(24)_m \times 10^{-7}$	$-9.8(2)_m(30)_{ex} \times 10^{-8}$
$V_{f+f}^{\prime\prime, \Sigma'}$	$-6.08(55)_m(2)_{ex} \times 10^{-7}$	$2.2(13)_m(1)_{ex} \times 10^{-8}$	$-4.0(41)_m(12)_{ex} \times 10^{-9}$
$V_{g+f}^{\prime\prime, \Sigma'}$	$6.67(44)_m(1)_{ex} \times 10^{-6}$	$-3.1(23)_m \times 10^{-7}$	$9.5(2)_m(29)_{ex} \times 10^{-8}$
$V_{g+f}^{\prime\prime, \Sigma'}$	$5.70(63)_m(2)_{ex} \times 10^{-7}$	$-2.0(12)_m(1)_{ex} \times 10^{-8}$	$3.9(39)_m(12)_{ex} \times 10^{-9}$
$V_{f+}^{\Sigma'}$		$5.3(22)_m \times 10^{-5}$	$-1.385(18)_m \times 10^{-5}$
$V_{f+}^{\Sigma'}$		$8.85(50)_m \times 10^{-6}$	$-4.9(47)_m \times 10^{-7}$
$V_{g+}^{\Sigma'}$	$4.61(27)_m \times 10^{-4}$	$-5.6(24)_m \times 10^{-5}$	$1.556(19)_m \times 10^{-5}$
$V_{g+}^{\Sigma'}$	$1.27(33)_m \times 10^{-4}$	$-9.37(53)_m \times 10^{-6}$	$5.5(53)_m \times 10^{-7}$
$V_{g+}^{\prime\prime, \Sigma'}$	$-1.236(48)_m(2)_{ex} \times 10^{-4}$	$-2.27(23)_m(2)_{ex} \times 10^{-5}$	$4.30(23)_m(50)_{ex} \times 10^{-6}$
$V_{g+}^{\prime\prime, \Sigma'}$	$3.6(22)_m \times 10^{-5}$	$-7.49(68)_m(10)_{ex} \times 10^{-6}$	$2.6(21)_m(3)_{ex} \times 10^{-7}$
$V_{f-}^{\Sigma'}$	$2.83(17)_m \times 10^{-4}$	$-3.6(16)_m \times 10^{-5}$	$9.83(14)_m \times 10^{-6}$
$V_{f-}^{\Sigma'}$	$7.7(20)_m \times 10^{-5}$	$-5.95(36)_m \times 10^{-6}$	$3.4(33)_m \times 10^{-7}$
$V_{f-}^{\prime\prime, \Sigma'}$	$-5.98(84)_m(1)_{ex} \times 10^{-4}$	$3.00(97)_m(1)_{ex} \times 10^{-4}$	$-9.8(2)_m(16)_{ex} \times 10^{-5}$
$V_{f-}^{\prime\prime, \Sigma'}$	$-3.0(10)_m \times 10^{-4}$	$6.93(13)_m(10)_{ex} \times 10^{-5}$	$-4.9(44)_m(8)_{ex} \times 10^{-6}$
$V_{g-}^{\Sigma'}$	$-3.01(18)_m \times 10^{-4}$	$3.9(17)_m \times 10^{-5}$	$-1.108(15)_m \times 10^{-5}$
$V_{g-}^{\Sigma'}$	$-8.2(21)_m \times 10^{-5}$	$6.55(38)_m \times 10^{-6}$	$-3.9(38)_m \times 10^{-7}$
$V_{ffilde}^{\Sigma'}$	$4.02(36)_m \times 10^{-6}$	$2.00(68)_m \times 10^{-6}$	$-7.9845(91)_m \times 10^{-7}$
$V_{ffilde}^{\Sigma'}$	$-1.68(92)_m \times 10^{-6}$	$4.119(44)_m \times 10^{-7}$	$-3.1(30)_m \times 10^{-8}$
$V_{ffilde}^{\prime\prime, \Sigma'}$	$-9.13(54)_m \times 10^{-6}$	$1.41(54)_m \times 10^{-6}$	$-3.64(3)_m(66)_{ex} \times 10^{-7}$
$V_{ffilde}^{\prime\prime, \Sigma'}$	$-3.29(98)_m \times 10^{-6}$	$2.71(8)_m(13)_{ex} \times 10^{-7}$	$-1.8(16)_m(4)_{ex} \times 10^{-8}$
$V_{gfilde}^{\Sigma'}$	$-3.78(34)_m \times 10^{-6}$	$-1.96(67)_m \times 10^{-6}$	$7.822(12)_m \times 10^{-7}$
$V_{gfilde}^{\Sigma'}$	$1.71(93)_m \times 10^{-6}$	$-4.015(50)_m \times 10^{-7}$	$3.1(29)_m \times 10^{-8}$
$V_{gfilde}^{\prime\prime, \Sigma'}$	$9.07(53)_m \times 10^{-6}$	$-1.35(52)_m \times 10^{-6}$	$3.49(3)_m(64)_{ex} \times 10^{-7}$
$V_{gfilde}^{\prime\prime, \Sigma'}$	$3.22(95)_m \times 10^{-6}$	$-2.57(8)_m(13)_{ex} \times 10^{-7}$	$1.7(15)_m(3)_{ex} \times 10^{-8}$
V_f^{Δ}	$1.183(60)_m \times 10^{-3}$	$-4.44(94)_m \times 10^{-5}$	$-1.28(34)_m \times 10^{-7}$
V_f^{Δ}	$3.31(32)_m \times 10^{-4}$	$-5.7(13)_m \times 10^{-6}$	$7.36(39)_m \times 10^{-8}$
V_g^{Δ}	$-1.228(62)_m \times 10^{-3}$	$4.8(10)_m \times 10^{-5}$	$1.43(40)_m \times 10^{-7}$
V_g^{Δ}	$-3.44(33)_m \times 10^{-4}$	$6.2(14)_m \times 10^{-6}$	$-8.53(47)_m \times 10^{-8}$
V_{f-f}^{Δ}		$4.51(99)_m \times 10^{-6}$	$1.22(50)_m \times 10^{-8}$
V_{f-f}^{Δ}		$5.9(13)_m \times 10^{-7}$	$-1.116(77)_m \times 10^{-8}$
V_{g-f}^{Δ}	$4.29(29)_m \times 10^{-5}$	$-4.35(95)_m \times 10^{-6}$	$-1.20(47)_m \times 10^{-8}$
V_{g-f}^{Δ}	$1.25(13)_m \times 10^{-5}$	$-5.7(12)_m \times 10^{-7}$	$1.072(73)_m \times 10^{-8}$
V_{f+f}^{Δ}	$-8.10(15)_m \times 10^{-6}$	$2.25(44)_m \times 10^{-7}$	$1.19(19)_m \times 10^{-9}$
V_{f+f}^{Δ}	$-2.09(16)_m \times 10^{-6}$	$2.79(77)_m \times 10^{-8}$	$-4.078(89)_m \times 10^{-10}$
V_{g+f}^{Δ}	$7.82(14)_m \times 10^{-6}$	$-2.15(42)_m \times 10^{-7}$	$-1.16(19)_m \times 10^{-9}$
V_{g+f}^{Δ}	$2.02(15)_m \times 10^{-6}$	$-2.67(73)_m \times 10^{-8}$	$3.925(80)_m \times 10^{-10}$